Online Appendix to "Revisiting Chart-Based Return Prediction: Shared Rankings and Economic Limits"

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Appendix A. Implementation and reporting conventions

This supplement records the implementation details and result grids that support the article. Every numerical cell is a direct transcription or a compact rearrangement of the finalized analysis underlying the article.

Appendix A.1. Sample, timing, and portfolio design

Table A.1 collects the choices that apply across the learned models. The chart CNN, CNN1D, and multimodal models use the same CRSP source panel, target definition, sample split, and portfolio rules. A method enters a formation-date cross section when its required input history and subsequent holding-period return are available.

Appendix A.2. Comparison with Jiang et al.'s design

Table A.2 separates inherited design elements from extensions and implementation differences. The comparison is descriptive: the analysis does not attribute differences in portfolio performance to any single choice.

Appendix A.3. Representations and network architectures

The chart input contains binary open-high-low-close bars, a moving-average line with length I, and a separate volume panel. Prices and the moving average are jointly scaled to the stock-window high–low range, and volume is scaled separately. The image-scaled sequence retains the six continuous open, high, low, close, moving-average, and volume channels after the corresponding within-window scaling. CNN1D applies convolution along the ordered trading-day dimension.

The scaling exercise also estimates a logistic classifier on the image-scaled, cumulative-return-scaled, and devolatized-return-scaled sequences. Scheme-specific sequence channels are standardized with moments estimated in the development sample and then held fixed during evaluation.

For multimodal feature fusion, let $f_{i,t}^{2D}$ and $f_{i,t}^{1D}$ denote the chart and sequence feature vectors. The joint representation is

$$f_{i,t}^{MM} = \left[f_{i,t}^{2D}; f_{i,t}^{1D}\right]. \tag{A.1}$$

Table A.1: Empirical design summary

Component	Convention
Source universe	CRSP ordinary common shares with share codes 10 and 11 on NYSE, AMEX, or Nasdaq. Prices must be positive, and lagged market capitalization must be observed for value-weighted portfolios.
Development sample	1993–2000. Stock-date observations are randomly divided into 70% training and 30% validation sets using a fixed seed.
Evaluation samples	Full period 2001–2024, baseline period 2001–2019, and extension period 2020–2024.
Input and target grid	Input length $I \in \{5, 20, 60\}$ trading days and future return horizon $R \in \{5, 20, 60\}$ trading days. The target equals one when the cumulative return over dates $t + 1$ through $t + R$ is positive.
Portfolio frequency	Non-overlapping weekly, monthly, and quarterly formation dates for $R = 5$, 20, and 60.
Portfolio rule	Buy the highest-score decile and short the lowest-score decile. Each leg has one dollar of notional. Results use equal- or lagged-market-capitalization weights within each leg.
Annualization	Mean returns use factors 52, 12, and 4. Standard deviations use the square roots of the same factors. The Sharpe ratio is annualized mean divided by annualized standard deviation.
Transaction costs	A one-way rate of 10 basis points applies above the contemporaneous NYSE 80th-percentile market-capitalization breakpoint and 20 basis points otherwise. The deduction uses full absolute traded notional in both legs.
Turnover	Conventional half-turnover is computed separately for the long and short legs, summed, and expressed as a monthly equivalent.

The concatenated vector enters a linear classifier. Late-fusion specifications instead average the probabilities or logits from separately trained branches. For every learned model, the softmax probability assigned to the up class is

$$s_{i,t} = \frac{\exp(z_{i,t}^{(1)})}{\exp(z_{i,t}^{(0)}) + \exp(z_{i,t}^{(1)})}. \tag{A.2}$$

This probability is the within-date portfolio ranking score.

Table A.2: Design comparison with Jiang et al. (2023)

Dimension	Jiang et al. (2023)	This study
Development sample	1993–2000; random 70% training and 30% validation split	1993–2000; random 70% training and 30% validation split using a fixed seed
Evaluation sample	2001–2019	2001–2024, reported as a 2001–2019 baseline and a 2020–2024 extension
Core model grid	Chart CNN, logistic classifier, and CNN1D; three scaling schemes and all nine (I, R) cells	Same core grid, plus multimodal feature fusion, late fusion, and direct learned-score comparisons
Chart-CNN forecast aggregation	Five independent fits per configuration with forecasts averaged	One selected checkpoint in the main analysis; matched one- and five-member sensitivity checks for $I5/R5$ and $I20/R5$
Optimizer and learning rate	Adam with initial learning rate 10^{-5}	Adam with learning rate 10^{-4}
Portfolio construction	Non-overlapping high-minus-low decile portfolios; equal and value weighting; 5-, 20-, and 60-day horizons	Same core portfolio construction, with exact common stock-date panels imposed for learned-score overlap tests
Price-based benchmarks	MOM, STR, WSTR, and TREND	Same four benchmarks under the common portfolio protocol
Reported cost schedule	10 basis points for stocks above the NYSE 80th-percentile size breakpoint and 20 basis points otherwise	Same rate schedule; costs deducted from full absolute traded notional in both portfolio legs
Tradability restrictions	Largest 500 stocks by market capitalization	Price of at least \$5; dollar volume of at least \$1 million or \$5 million; exclusion of NYSE microcaps; and a combined screen
Recent-period analysis	Evaluation ends in 2019	Separate 2020–2024 results with model parameters and portfolio rules held fixed

Note. The Jiang et al. entries summarize their Sections II.C, III, and IV.B, Tables IV and IX, and Internet Appendix Table IA.IV. “Same” denotes an inherited design element, not an independently introduced feature. Differences in results can reflect several choices jointly as well as the additional evaluation years.

Table A.3: Models and scaling schemes in the representation comparison

Model	Scaling	Input representation	Model structure
Logistic classifier	Cumulative-return scale	Continuous sequence	Linear logit
Logistic classifier	Devolatized-return scale	Continuous sequence	Linear logit
Logistic classifier	Image scale	Continuous sequence	Linear logit
CNN1D	Cumulative-return scale	Continuous sequence	Temporal convolution
CNN1D	Devolatized-return scale	Continuous sequence	Temporal convolution
CNN1D	Image scale	Continuous sequence	Temporal convolution
Chart CNN	Image scale	Binary grayscale image	Two-dimensional convolution
Multimodal model	Image scale	Chart and sequence	Paired branches with fusion

Table A.4: Architecture by input length

I	Chart input	Chart CNN blocks and channels	Sequence input	CNN1D blocks and channels
5	32×15	2 blocks, 64 and 128	6×5	1 block, 64
20	64×60	3 blocks, 64, 128, and 256	6×20	2 blocks, 64 and 128
60	96×180	4 blocks, 64, 128, 256, and 512	6×60	3 blocks, 64, 128, and 256

Note. Every chart block applies a 5×3 convolution, batch normalization, Leaky ReLU activation, and max pooling. Every sequence block applies the corresponding operations with a one-dimensional filter of length three, stride one, and padding one. The final feature map is flattened, dropout is applied, and a fully connected layer produces two return-sign logits. The multimodal model pairs the chart and sequence branches for the same (i, t, I, R) observation. Its primary fusion head concatenates the two branch feature vectors and applies a linear classifier.

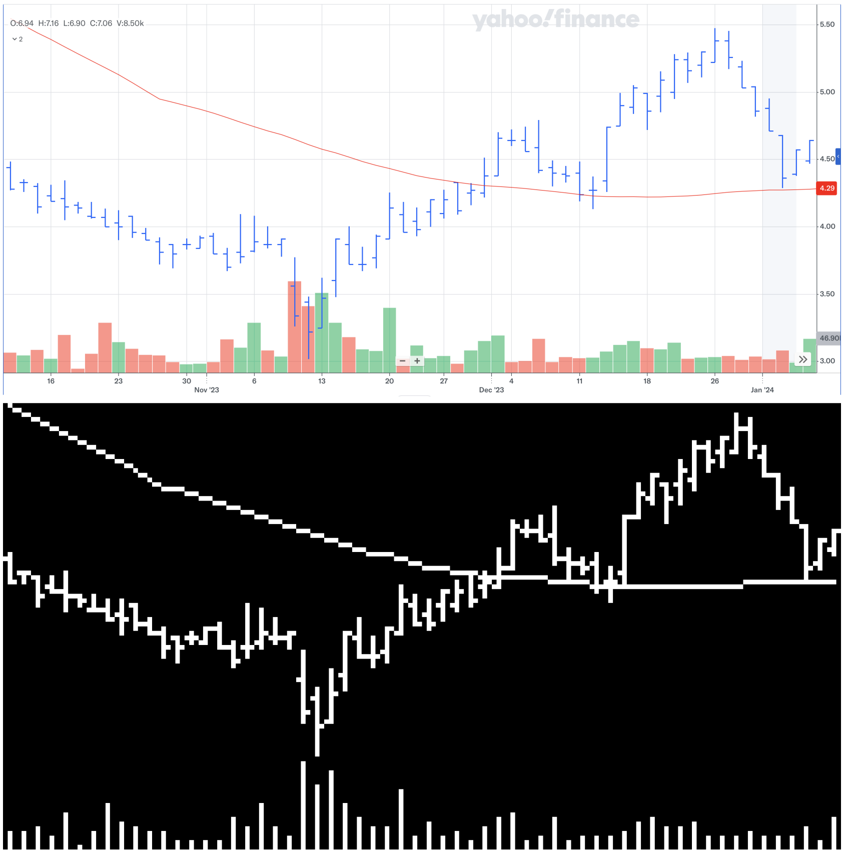

Figure A.1: Example of chart construction and model input. The upper panel shows a conventional 60-day OHLC chart for Envela Corporation (ELA). The lower panel shows the corresponding binary grayscale image used by the chart CNN, with the OHLC bars and moving-average line in the price panel and trading volume in the lower panel.

Appendix A.4. Training, checkpoint, and ensemble conventions

Table A.5 records the settings shared by the learned models. The main empirical analysis uses one selected checkpoint for every learned specification. The chart CNN seed sensitivity exercise separately compares one- and five-member forecast averages for the two short-input weekly specifications. Table A.6 keeps the main results and this sensitivity check distinct.

Appendix A.5. Benchmark and transaction-cost formulas

The article compares the learned rankings with four transparent price-based benchmarks. MOM is the cumulative return over the previous 252 trading days after excluding the most recent 21 trading days,

$$z_{i,t}^{\mathrm{MOM}} = \prod_{j=22}^{273}(1 + r_{i,t-j}) - 1. \tag{A.3}$$

Table A.5: Estimation settings for learned models

Setting	Value
Objective	Two-class cross-entropy loss
Optimizer	Adam
Learning rate	10^{-4}
Batch size	128 observations
Maximum epochs	50
Checkpoint rule	Lowest validation loss
Early stopping	Two consecutive epochs without validation-loss improvement
Dropout probability	0.50 after the flattened feature representation
Initialization	Xavier initialization for convolutional and fully connected weights
Portfolio score	Softmax probability assigned to the positive-return class

Table A.6: Run convention by result family

Result family	Forecast aggregation	Scope
Weekly mechanism grid	One selected checkpoint	All chart CNN, CNN1D, and multimodal $I5/R5$, $I20/R5$, and $I60/R5$ cells
Chart grid	One selected checkpoint	All nine (I, R) cells
CNN1D grid	One selected checkpoint	All nine (I, R) cells
Multimodal grid	One selected checkpoint	All nine feature-concatenation cells
Seed sensitivity	Matched $M = 1$ and $M = 5$ results	Chart CNN $I5/R5$ and $I20/R5$
Late fusion	Prediction-level combination	Pretrained chart and CNN1D probabilities or logits are averaged after branch prediction

One-month short-term reversal (STR) and one-week short-term reversal (WSTR) use the negative of the previous 21- and 5-trading-day returns,

$$z_{i,t}^{\mathrm{STR}} = -\left(\frac{P_{i,t}}{P_{i,t-21}} - 1\right), \qquad z_{i,t}^{\mathrm{WSTR}} = -\left(\frac{P_{i,t}}{P_{i,t-5}} - 1\right). \tag{A.4}$$

High values therefore identify recent losers. The price series is split-consistent with the return history.

TREND combines moving-average information for

$$\mathcal{L} = \{3, 5, 10, 20, 50, 100, 200, 400, 600, 800, 1000\}.$$

For each L, define $x_{i,t,L} = \mathrm{MA}_{i,t,L}/P_{i,t}$. A date-level cross-sectional regression maps these ratios into next-period returns, and the forecasting coefficients are the trailing mean of the previous 12 fully realized coefficient vectors. The score is

$$z_{i,t}^{\mathrm{TREND}} = \sum_{L \in \mathcal{L}} \overline{\beta}_{L,t} x_{i,t,L}. \tag{A.5}$$

For $\ell \in \{\text{long}, \text{short}\}$, conventional half-turnover is

$$\mathrm{Turnover}_t^{\ell} = \frac{1}{2} \sum_i \left| w_{i,t}^{\ell,\mathrm{target}} - w_{i,t}^{\ell,\mathrm{prev}} \right|. \tag{A.6}$$

Reported turnover sums the two legs and converts the average to a monthly equivalent. Net performance deducts the one-way cost rate $c_{i,t}$ from full absolute traded notional in each leg,

$$R^{\mathrm{net}}_{p,t} = R^{\mathrm{gross}}_{p,t} - \sum_{i,\ell} c_{i,t} \left|\Delta w^{\ell}_{i,t}\right|. \tag{A.7}$$

The rate is 10 basis points above the contemporaneous NYSE 80th-percentile market-capitalization breakpoint and 20 basis points otherwise.

Appendix B. Complete representation decomposition

Table B.7 reports the full scaling exercise from the main analysis. It extends the article's weekly main grid to monthly and quarterly return horizons while preserving both portfolio weighting rules.

Table B.7: Gross Sharpe ratios across sequence scaling schemes

	5-day input		20-day input		60-day input	
Model	EW	VW	EW	VW	EW	VW
Panel A. 5-day return horizon						
Chart CNN	5.88	1.39	6.52	1.44	3.72	1.18
Logistic, image scale	4.88	1.25	5.22	1.09	5.44	1.12
Logistic, cumulative scale	0.18	−0.04	−0.41	−0.51	−0.26	−0.53
Logistic, devolatized scale	2.80	0.79	2.73	0.48	2.70	0.66
CNN1D, image scale	6.24	1.16	6.55	1.12	5.89	2.06
CNN1D, cumulative scale	4.88	1.47	5.12	1.41	4.66	1.15
CNN1D, devolatized scale	4.39	0.73	5.26	1.04	4.82	1.28
Panel B. 20-day return horizon						
Chart CNN	1.35	0.10	2.26	0.49	0.57	−0.29
Logistic, image scale	2.03	0.56	2.37	0.41	1.59	0.60
Logistic, cumulative scale	0.02	0.37	−0.29	−0.08	−0.30	−0.64
Logistic, devolatized scale	1.27	0.59	1.02	0.28	0.98	0.12
CNN1D, image scale	2.18	0.41	2.15	0.41	1.74	0.51
CNN1D, cumulative scale	0.94	0.92	0.97	0.56	1.45	0.63
CNN1D, devolatized scale	1.17	0.16	1.73	0.53	1.22	0.30
Panel C. 60-day return horizon						
Chart CNN	0.23	−0.18	0.34	0.21	0.55	0.45
Logistic, image scale	0.97	0.04	0.94	0.21	0.55	0.21
Logistic, cumulative scale	0.37	0.11	−0.02	−0.06	−0.30	−0.54
Logistic, devolatized scale	0.66	0.08	0.61	0.37	0.40	0.17
CNN1D, image scale	0.99	0.28	0.46	0.15	0.46	0.25
CNN1D, cumulative scale	0.26	0.33	0.13	0.16	0.34	0.21
CNN1D, devolatized scale	0.83	0.13	0.65	0.34	0.45	0.21

Note. Entries are annualized gross Sharpe ratios of high-minus-low decile portfolios over 2001–2024. EW and VW denote equal- and value-weighted legs. Image scale applies the chart's within-window high–low normalization to the continuous sequence. Cumulative scale anchors the path to the first close, and devolatized scale expresses changes relative to lagged volatility.

Appendix C. Complete learned-model grids

Tables C.8–C.10 condense the baseline and extension results into complete 3×3 grids. Gross SR and Net SR denote annualized Sharpe ratios before and after the common transaction-cost deduction. This layout keeps horizon, sample period, and weighting visible without repeating the return and standard-deviation columns already summarized in the article's focused economic tests.

Table C.8: Chart CNN Sharpe-ratio grid

	Equal-weighted				Value-weighted			
	2001–2019		2020–2024		2001–2019		2020–2024	
Spec.	Gross SR	Net SR	Gross SR	Net SR	Gross SR	Net SR	Gross SR	Net SR
$I5/R5$	7.00	4.28	3.30	1.01	1.77	0.00	0.57	−0.47
$I5/R20$	1.98	1.05	0.21	−0.37	0.26	−0.23	−0.16	−0.41
$I5/R60$	0.14	−0.10	0.56	0.28	−0.15	−0.28	0.14	−0.03
$I20/R5$	7.64	4.78	3.87	1.35	1.71	−0.01	0.70	−0.62
$I20/R20$	2.53	1.51	1.76	1.30	0.26	−0.31	0.67	0.54
$I20/R60$	0.35	0.16	0.02	−0.15	0.25	0.08	0.12	−0.01
$I60/R5$	5.32	2.26	1.18	−0.61	1.27	−0.68	1.13	−0.45
$I60/R20$	1.28	0.31	−0.51	−0.98	−0.41	−0.97	−0.22	−0.51
$I60/R60$	0.25	0.07	0.46	0.36	0.05	−0.01	0.74	0.53

Note. Each cell reports one selected chart CNN specification from the main analysis. Returns are formed at the frequency associated with R and are annualized using 52, 12, or 4 observations per year.

Table C.9: CNN1D Sharpe-ratio grid

	Equal-weighted				Value-weighted			
	2001–2019		2020–2024		2001–2019		2020–2024	
Spec.	Gross SR	Net SR	Gross SR	Net SR	Gross SR	Net SR	Gross SR	Net SR
$I5/R5$	7.45	4.50	3.47	1.02	1.37	−0.21	0.54	−0.49
$I5/R20$	2.78	1.78	1.26	0.67	0.46	0.00	0.52	0.19
$I5/R60$	1.08	0.83	0.73	0.51	0.44	0.28	−0.07	−0.17
$I20/R5$	8.09	5.28	3.34	1.18	1.49	−0.35	0.28	−0.77
$I20/R20$	2.63	1.92	1.30	0.85	0.38	−0.02	0.70	0.25
$I20/R60$	0.49	0.32	0.37	0.24	0.32	0.19	−0.31	−0.45
$I60/R5$	7.88	5.58	2.29	0.78	2.61	0.80	0.92	−0.29
$I60/R20$	2.07	1.59	1.21	0.92	0.60	0.26	0.29	−0.02
$I60/R60$	0.49	0.37	0.44	0.32	0.36	0.23	0.06	−0.03

Note. Every cell uses one selected checkpoint of the image-scaled CNN1D model. Gross and net Sharpe ratios follow the portfolio and transaction-cost conventions in Table A.1.

Table C.10: Multimodal Sharpe-ratio grid

	Equal-weighted				Value-weighted			
	2001–2019		2020–2024		2001–2019		2020–2024	
Spec.	Gross SR	Net SR	Gross SR	Net SR	Gross SR	Net SR	Gross SR	Net SR
$I5/R5$	7.23	4.41	3.26	0.97	1.53	−0.15	0.25	−0.86
$I5/R20$	1.81	0.83	0.31	−0.24	0.11	−0.36	0.39	0.07
$I5/R60$	0.74	0.46	0.27	−0.04	0.08	−0.07	−0.25	−0.39
$I20/R5$	7.76	5.12	3.19	1.11	1.74	−0.01	0.90	−0.35
$I20/R20$	2.51	1.68	1.56	1.04	0.41	−0.02	0.02	−0.38
$I20/R60$	0.60	0.40	0.38	0.25	0.05	−0.12	0.19	0.08
$I60/R5$	7.58	5.31	1.91	0.44	2.23	0.44	0.64	−0.75
$I60/R20$	1.89	1.31	−0.14	−0.40	0.75	0.35	0.02	−0.35
$I60/R60$	0.75	0.57	0.53	0.43	0.43	0.29	0.64	0.50

Note. Every cell uses one selected feature-concatenation checkpoint. Each observation pairs chart and sequence inputs for the same stock, formation date, input length, and forecast horizon.

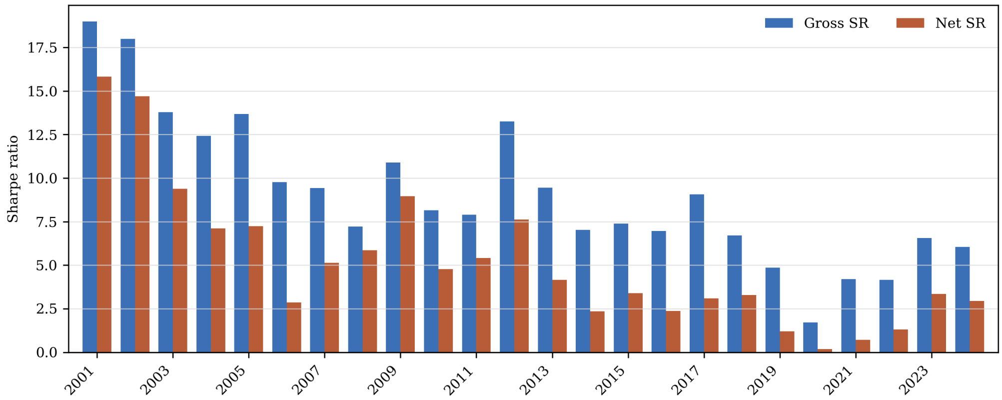

Figure C.2: Annual gross and net Sharpe ratios for the representative chart CNN

Note. The figure reports annual Sharpe ratios for the equal-weighted chart CNN $I20/R5$ portfolio from 2001 through 2024. The net series applies the common transaction-cost schedule.

Appendix D. Rank stability across input windows

Table D.11 reports the underlying correlations in a compact tabular form. Each row holds the model family and return horizon fixed and compares the cross-sectional score rankings produced by two input lengths.

Table D.11: Within-model Spearman rank correlations across input windows

Model	Return horizon	$I5/I20$	$I5/I60$	$I20/I60$
Chart CNN	$R5$	0.60	0.22	0.34
Chart CNN	$R20$	0.28	0.07	0.09
Chart CNN	$R60$	0.18	0.09	0.33
CNN1D	$R5$	0.64	0.54	0.71
CNN1D	$R20$	0.52	0.38	0.62
CNN1D	$R60$	0.30	0.21	0.45
Multimodal	$R5$	0.56	0.44	0.59
Multimodal	$R20$	0.23	0.19	0.37
Multimodal	$R60$	0.21	0.11	0.32

Note. Entries are time-series averages of formation-date Spearman correlations between two input-window scores from the same learned-model family over 2001–2024. The correlations are computed on the common stock-date panel used in the main overlap analysis.

Appendix E. Seed and fusion sensitivity

The chart seed comparison is available for the two short-input weekly specifications. Table E.12 shows that the matched one- and five-member results are close in these cells. Table E.13 compares probability and logit averaging using the same aligned branch predictions.

Table E.12: Matched one- and five-member chart CNN results

Spec.	Weight	Gross SR, $M = 1$	Gross SR, $M = 5$	Net return, $M = 1$	Net return, $M = 5$	Net SR, $M = 1$	Net SR, $M = 5$
$I5/R5$	EW	5.88	5.99	48.8%	49.9%	3.44	3.54
$I5/R5$	VW	1.39	1.44	−2.1%	−1.2%	−0.13	−0.07
$I20/R5$	EW	6.52	6.58	51.3%	55.3%	3.94	4.10
$I20/R5$	VW	1.44	1.40	−2.5%	−2.2%	−0.16	−0.13

Note. The table reports full-sample 2001–2024 portfolios under matched construction rules. M is the number of independently trained chart forecasts averaged before the cross-sectional sort.

Table E.13: Late-fusion sensitivity to probability and logit averaging

Spec.	Fusion rule	EW full	EW baseline	EW extension	VW full	VW baseline	VW extension
$I5/R5$	Probability average	6.25	7.26	3.47	1.21	1.41	0.55
$I5/R5$	Logit average	6.25	7.25	3.47	1.20	1.40	0.55
$I20/R5$	Probability average	6.51	7.70	3.40	1.60	1.89	0.82
$I20/R5$	Logit average	6.51	7.69	3.39	1.59	1.86	0.84

Note. Entries are annualized gross Sharpe ratios. Baseline and extension denote 2001–2019 and 2020–2024. Late fusion combines the aligned predictions of pretrained chart and sequence branches.

Appendix F. Price-based benchmark grids

The benchmark tables use the same portfolio frequencies, weighting rules, annualization, and transaction-cost schedule as the learned portfolios. MOM is the implemented medium-horizon momentum signal. It compounds returns over trading-day lags 22 through 273 and excludes the most recent 21 trading days. STR and WSTR are one-month and one-week reversal signals. TREND is the supervised rolling linear forecast built from moving-average-to-price ratios.

Table F.14: Equal-weighted price-based benchmark Sharpe ratios

		2001–2019		2020–2024		2001–2024	
Frequency	Signal	Gross SR	Net SR	Gross SR	Net SR	Gross SR	Net SR
Week	MOM	−0.19	−0.36	0.26	0.12	−0.08	−0.25
Week	STR	2.04	1.33	1.10	0.44	1.83	1.13
Week	WSTR	3.33	1.59	1.42	−0.03	2.86	1.20
Week	TREND	3.09	1.85	0.38	−0.62	2.49	1.31
Month	MOM	−0.01	−0.09	0.34	0.27	0.06	−0.01
Month	STR	0.71	0.32	0.24	−0.07	0.59	0.22
Month	WSTR	1.43	0.87	0.47	0.13	1.12	0.63
Month	TREND	0.70	0.37	0.60	0.30	0.68	0.35
Quarter	MOM	−0.13	−0.16	0.38	0.35	−0.04	−0.07
Quarter	STR	0.34	0.22	0.09	−0.00	0.28	0.17
Quarter	WSTR	0.37	0.18	0.56	0.30	0.39	0.19
Quarter	TREND	−0.05	−0.11	0.54	0.43	0.01	−0.06

Note. Gross and net Sharpe ratios are annualized. Net returns apply the common transaction-cost schedule to full absolute traded notional.

Table F.15: Value-weighted price-based benchmark Sharpe ratios

		2001–2019		2020–2024		2001–2024	
Frequency	Signal	Gross SR	Net SR	Gross SR	Net SR	Gross SR	Net SR
Week	MOM	0.12	−0.05	0.53	0.38	0.21	0.05
Week	STR	0.59	0.01	−0.51	−1.01	0.34	−0.23
Week	WSTR	0.82	−0.27	−0.59	−1.41	0.47	−0.55
Week	TREND	0.72	−0.09	−0.21	−0.77	0.52	−0.24
Month	MOM	0.18	0.11	0.26	0.21	0.20	0.13
Month	STR	0.24	−0.05	−0.31	−0.49	0.09	−0.17
Month	WSTR	0.32	−0.01	−0.25	−0.46	0.16	−0.13
Month	TREND	−0.11	−0.33	0.39	0.24	0.02	−0.19
Quarter	MOM	0.07	0.03	0.52	0.48	0.16	0.12
Quarter	STR	0.20	0.10	0.04	−0.02	0.15	0.07
Quarter	WSTR	−0.02	−0.16	−0.07	−0.14	−0.04	−0.15
Quarter	TREND	−0.03	−0.08	1.21	1.13	0.13	0.07

Note. Gross and net Sharpe ratios are annualized. Value weights use lagged market capitalization and are normalized separately within the long and short legs.

Appendix G. Decile profiles

The complete decile profiles show whether the high-minus-low result is supported by a broader cross-sectional ordering. The six panels report annualized gross decile returns for each model, input length, forecast horizon, and weighting rule over 2001–2024. They reproduce the return panels from the finalized simple-table exhibits and use no graphical interpolation. The full-grid Sharpe ratios appear in Tables C.8–C.10.

Table G.16: Weekly learned-model decile performance, equal-weighted portfolios

Annualized gross returns

	Chart CNN			CNN1D			Multimodal		
Decile	*I5/R5*	*I20/R5*	*I60/R5*	*I5/R5*	*I20/R5*	*I60/R5*	*I5/R5*	*I20/R5*	*I60/R5*
Low	-27.1%	-31.9%	-16.4%	-26.5%	-34.6%	-38.7%	-27.0%	-34.3%	-35.2%
2	-3.8%	-2.8%	2.2%	-3.5%	-4.7%	-3.7%	-5.0%	-4.4%	-4.0%
3	2.9%	4.8%	9.4%	1.7%	4.9%	5.9%	1.8%	4.5%	6.9%
4	6.9%	9.6%	12.5%	5.5%	9.5%	11.4%	6.7%	9.0%	10.9%
5	10.3%	12.4%	15.0%	9.5%	13.6%	15.8%	10.9%	12.9%	14.6%
6	13.8%	15.6%	16.8%	12.4%	16.6%	19.2%	13.2%	16.9%	17.8%
7	18.1%	19.4%	18.8%	17.5%	20.1%	21.4%	18.6%	20.6%	21.0%
8	23.5%	23.0%	21.7%	25.0%	23.7%	25.1%	24.7%	23.3%	24.3%
9	32.1%	29.5%	23.4%	34.8%	29.4%	29.1%	31.8%	31.1%	29.4%
High	56.1%	53.0%	29.7%	56.4%	54.5%	49.5%	57.0%	53.3%	49.3%
H–L	83.3%	84.9%	46.1%	82.9%	89.1%	88.2%	84.0%	87.5%	84.5%

Note. The table reports annualized gross decile returns over the available weekly observations in 2001–2024. Low is the lowest-score decile, High is the highest-score decile, and H–L is the long-short spread.

Table G.17: Weekly learned-model decile performance, value-weighted portfolios

Annualized gross returns

	Chart CNN			CNN1D			Multimodal		
Decile	*I5/R5*	*I20/R5*	*I60/R5*	*I5/R5*	*I20/R5*	*I60/R5*	*I5/R5*	*I20/R5*	*I60/R5*
Low	0.1%	-2.7%	-0.8%	1.9%	-0.2%	-11.9%	1.9%	-3.4%	-5.6%
2	5.3%	9.2%	4.8%	5.3%	4.5%	-0.0%	3.0%	3.0%	5.4%
3	4.8%	9.0%	6.6%	3.2%	6.5%	5.0%	8.0%	7.5%	5.0%
4	6.6%	6.6%	8.7%	7.1%	7.6%	8.0%	7.0%	8.4%	6.1%
5	9.4%	8.1%	6.9%	7.9%	7.6%	6.6%	9.0%	7.2%	8.3%
6	9.5%	7.8%	11.6%	10.3%	10.5%	7.9%	9.6%	9.8%	8.2%
7	10.0%	9.9%	9.9%	10.7%	10.5%	8.5%	11.5%	10.7%	9.1%
8	13.3%	13.3%	11.2%	17.4%	11.2%	12.4%	12.3%	11.7%	12.0%
9	14.6%	14.1%	12.4%	18.1%	15.6%	13.2%	16.9%	14.9%	13.1%
High	23.9%	20.5%	13.9%	23.2%	17.9%	20.0%	21.8%	20.2%	19.3%
H–L	23.8%	23.2%	14.7%	21.3%	18.0%	31.9%	19.9%	23.6%	25.0%

Note. The table reports annualized gross decile returns over the available weekly observations in 2001–2024. Low is the lowest-score decile, High is the highest-score decile, and H–L is the long-short spread.

Table G.18: Monthly learned-model decile performance, equal-weighted portfolios

Annualized gross returns

	Chart CNN			CNN1D			Multimodal		
Decile	$I5/R20$	$I20/R20$	$I60/R20$	$I5/R20$	$I20/R20$	$I60/R20$	$I5/R20$	$I20/R20$	$I60/R20$
Low	-3.2%	-2.1%	0.7%	-9.2%	-5.8%	-9.7%	-3.2%	-3.1%	-8.4%
2	0.3%	5.1%	2.2%	-2.2%	4.4%	4.2%	-0.3%	5.6%	-1.0%
3	2.2%	7.7%	2.9%	-0.3%	9.6%	7.8%	0.6%	9.2%	1.2%
4	1.7%	10.2%	2.5%	1.5%	10.5%	11.5%	-0.4%	10.1%	2.8%
5	3.8%	12.7%	2.8%	0.9%	12.5%	13.8%	0.9%	11.2%	3.9%
6	3.2%	12.1%	2.5%	1.6%	13.6%	16.1%	1.4%	13.4%	5.8%
7	3.3%	14.2%	4.1%	4.4%	13.6%	17.1%	2.7%	14.1%	6.6%
8	4.8%	13.8%	4.1%	4.6%	16.0%	17.0%	1.8%	14.9%	6.2%
9	5.4%	17.7%	4.2%	5.6%	17.2%	17.8%	5.8%	16.4%	7.7%
High	10.6%	21.8%	6.7%	11.9%	22.0%	20.2%	9.5%	21.8%	10.3%
H–L	13.8%	24.0%	6.0%	21.0%	27.8%	29.8%	12.7%	24.9%	18.7%

Note. The table reports annualized gross decile returns over the available monthly observations in 2001–2024. Low is the lowest-score decile, High is the highest-score decile, and H–L is the long-short spread.

Table G.19: Monthly learned-model decile performance, value-weighted portfolios

Annualized gross returns

	Chart CNN			CNN1D			Multimodal		
Decile	$I5/R20$	$I20/R20$	$I60/R20$	$I5/R20$	$I20/R20$	$I60/R20$	$I5/R20$	$I20/R20$	$I60/R20$
Low	1.2%	6.4%	4.5%	1.1%	5.5%	4.0%	3.0%	6.1%	-5.1%
2	0.6%	5.4%	2.7%	0.2%	8.3%	4.6%	1.0%	6.5%	-0.8%
3	1.2%	7.6%	1.2%	-0.2%	8.7%	8.6%	1.4%	10.6%	-3.3%
4	-1.1%	7.3%	3.8%	1.5%	8.8%	6.7%	2.9%	9.1%	3.4%
5	3.0%	7.8%	3.4%	1.7%	9.7%	9.1%	0.2%	9.2%	1.2%
6	0.1%	8.7%	-1.0%	2.9%	10.5%	9.3%	1.8%	10.6%	1.4%
7	4.3%	10.8%	2.0%	0.9%	8.7%	12.0%	3.6%	9.6%	3.4%
8	3.1%	8.8%	0.4%	4.1%	8.9%	10.2%	2.2%	8.9%	2.0%
9	1.5%	12.3%	-0.3%	3.4%	11.1%	10.6%	1.7%	10.1%	3.0%
High	2.6%	12.4%	1.0%	7.0%	11.6%	12.4%	5.0%	10.5%	4.0%
H–L	1.4%	6.0%	-3.5%	5.9%	6.1%	8.4%	2.0%	4.4%	9.1%

Note. The table reports annualized gross decile returns over the available monthly observations in 2001–2024. Low is the lowest-score decile, High is the highest-score decile, and H–L is the long-short spread.

Table G.20: Quarterly learned-model decile performance, equal-weighted portfolios

Annualized gross returns

	Chart CNN			CNN1D			Multimodal		
Decile	$I5/R60$	$I20/R60$	$I60/R60$	$I5/R60$	$I20/R60$	$I60/R60$	$I5/R60$	$I20/R60$	$I60/R60$
Low	6.6%	5.0%	1.4%	7.0%	8.0%	7.3%	5.9%	1.9%	0.1%
2	6.2%	5.5%	5.0%	10.9%	10.7%	10.2%	6.4%	5.7%	3.1%
3	7.2%	6.4%	7.2%	11.0%	12.0%	11.1%	4.7%	7.9%	6.7%
4	6.2%	7.3%	8.0%	11.6%	12.6%	12.5%	6.5%	7.0%	8.1%
5	8.5%	7.5%	8.4%	13.1%	13.7%	13.4%	6.3%	7.9%	9.5%
6	8.3%	9.0%	8.7%	13.2%	12.7%	14.6%	7.6%	9.0%	9.2%
7	7.8%	7.9%	8.6%	13.8%	14.5%	14.4%	9.4%	8.0%	9.7%
8	8.8%	9.0%	10.5%	13.6%	13.4%	15.1%	9.2%	10.0%	10.7%
9	9.0%	10.7%	9.9%	16.0%	15.5%	16.2%	9.9%	10.7%	11.8%
High	9.1%	9.8%	11.1%	18.2%	15.7%	17.1%	11.7%	10.0%	12.3%
H–L	2.5%	4.8%	9.7%	11.2%	7.7%	9.8%	5.8%	8.1%	12.2%

Note. The table reports annualized gross decile returns over the available quarterly observations in 2001–2024. Low is the lowest-score decile, High is the highest-score decile, and H–L is the long-short spread.

Table G.21: Quarterly learned-model decile performance, value-weighted portfolios

Annualized gross returns

	Chart CNN			CNN1D			Multimodal		
Decile	$I5/R60$	$I20/R60$	$I60/R60$	$I5/R60$	$I20/R60$	$I60/R60$	$I5/R60$	$I20/R60$	$I60/R60$
Low	11.0%	5.4%	2.8%	7.7%	10.5%	9.9%	9.6%	7.8%	1.8%
2	9.4%	4.7%	6.1%	10.4%	8.9%	10.4%	9.9%	3.3%	4.4%
3	8.6%	7.1%	8.5%	9.1%	10.3%	10.6%	5.3%	9.0%	6.5%
4	9.3%	11.4%	6.8%	12.1%	12.8%	10.0%	6.2%	7.8%	8.5%
5	4.8%	6.8%	6.0%	12.5%	11.1%	10.0%	5.6%	9.0%	6.3%
6	7.8%	5.8%	6.2%	10.4%	9.8%	10.9%	6.6%	9.0%	8.3%
7	9.0%	10.6%	8.0%	10.9%	10.5%	10.7%	8.4%	7.4%	5.4%
8	6.5%	8.6%	8.1%	10.9%	9.6%	11.1%	9.5%	7.0%	9.3%
9	8.8%	6.7%	8.5%	9.9%	10.8%	10.5%	10.7%	7.7%	11.0%
High	8.5%	7.8%	9.0%	11.5%	12.6%	13.8%	8.9%	7.8%	8.8%
H–L	-2.5%	2.4%	6.2%	3.7%	2.1%	3.9%	-0.7%	0.1%	7.0%

Note. The table reports annualized gross decile returns over the available quarterly observations in 2001–2024. Low is the lowest-score decile, High is the highest-score decile, and H–L is the long-short spread.